

ST LOUIS LAMBERT, MO, USA

WMO: 724340

Lat: 38.753N

Lon: 90.374W

Elev: 162

StdP: 99.40

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13994

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-11.0	-20.4	0.6	-12.5	-17.7	0.8	-9.8	11.7	-0.3	10.7	0.9	4.2	300	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	35.6	24.9	34.2	24.5	32.9	23.9	26.4	32.8	25.7	31.9	25.0	31.1	4.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	20.0	30.0	23.8	19.0	29.6	23.1	18.2	28.8	82.9	32.7	79.8	31.9	76.8	31.0	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.9	8.0	DB	-17.1	38.0	3.2	1.8	-19.4	39.3	-21.2	40.3	-23.0	41.3	-25.3	42.6	(4)
(5)			WB	-17.8	27.5	3.1	0.9	-20.0	28.2	-21.8	28.7	-23.5	29.2	-25.7	29.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	0.2	2.6	8.2	14.4	19.8	24.8	26.9	26.1	21.9	15.3	8.3	2.6
	DBStd	10.63	6.60	6.47	6.24	5.34	4.56	3.53	3.20	3.02	4.18	5.17	5.57	5.84
	HDD10.0	1002	311	218	108	18	1	0	0	0	0	12	97	238
	HDD18.3	2433	563	440	317	141	39	3	0	0	15	123	303	488
	CDD10.0	2577	6	12	54	150	306	443	522	499	356	175	46	8
	CDD18.3	964	0	0	4	22	86	196	264	241	121	29	2	0
	CDH23.3	9853	0	2	36	196	724	1992	3043	2511	1111	226	13	0
CDH26.7	4101	0	0	4	46	222	829	1441	1092	410	55	1	0	
Wind	WSAvg	3.9	4.4	4.4	4.6	4.7	4.0	3.6	3.3	3.2	3.1	3.7	4.1	4.1
Precipitation	PrecAvg	1057	62	57	80	116	126	113	96	86	83	82	89	66
	PrecMax	1475	214	127	175	279	281	293	202	154	259	332	188	281
	PrecMin	740	18	11	29	47	26	14	19	23	11	16	17	11
	PrecStd	201	42	30	33	55	67	67	48	35	54	60	48	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	22.8	26.9	30.5	32.8	35.6	38.7	37.3	34.6	30.8	25.2	20.2
		MCWB	14.5	15.1	16.9	19.5	22.2	23.9	24.3	24.8	23.6	21.6	18.4	15.8
	2%	DB	16.0	19.1	23.9	27.9	30.8	34.0	36.1	35.3	32.7	28.2	21.9	16.9
		MCWB	12.1	13.2	15.8	18.2	21.7	24.2	25.2	24.8	23.1	19.9	15.3	13.4
	5%	DB	13.0	16.0	21.0	25.6	29.3	32.7	34.4	33.6	30.9	25.9	19.4	14.0
		MCWB	9.4	11.1	14.5	17.2	21.0	23.7	25.0	24.4	22.2	18.7	14.4	10.4
	10%	DB	9.7	12.9	18.2	23.3	27.5	31.3	32.9	31.9	29.0	23.6	17.2	11.2
		MCWB	6.3	9.0	12.9	16.2	20.0	23.0	24.4	23.6	21.2	17.6	13.1	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	16.7	18.8	21.1	24.5	26.3	27.4	27.4	25.3	22.9	19.1	16.8
		MCDB	18.8	20.4	24.9	27.8	30.0	32.2	33.9	33.7	31.5	28.7	23.7	19.8
	2%	WB	13.2	14.5	17.0	19.7	23.0	25.4	26.5	26.4	24.2	21.6	17.0	14.0
		MCDB	15.2	18.1	21.9	25.1	28.6	31.8	33.3	32.4	30.0	26.2	19.9	16.3
	5%	WB	9.8	11.8	15.4	18.5	22.1	24.6	25.8	25.5	23.5	20.2	15.4	11.0
		MCDB	12.8	15.1	20.3	23.7	27.4	30.8	32.3	31.3	28.9	24.3	18.7	13.0
	10%	WB	6.5	8.7	13.5	17.2	21.1	23.8	25.2	24.7	22.6	18.4	13.3	8.2
		MCDB	9.4	12.9	17.7	22.4	26.2	29.7	31.5	30.3	27.6	22.8	17.0	10.6
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.2	10.2	10.7	9.8	9.4	9.4	9.5	10.3	10.2	9.3	8.2
		MCDBR	13.3	14.7	13.7	13.5	11.2	10.7	10.6	11.1	11.4	12.2	12.5	12.0
		MCWBR	9.8	10.1	7.8	7.2	5.2	4.1	3.7	3.9	4.4	5.9	8.1	8.8
	5% WB	MCDBR	12.5	13.4	11.9	11.7	10.1	9.8	9.7	9.9	9.8	10.4	11.0	11.0
		MCWBR	10.2	10.2	7.7	7.1	5.3	4.4	4.1	4.2	4.3	6.0	8.4	9.3
Clear-Sky Solar Irradiance	taub	0.303	0.314	0.346	0.401	0.439	0.447	0.475	0.464	0.409	0.350	0.309	0.301	
	taud	2.474	2.437	2.412	2.253	2.210	2.222	2.143	2.173	2.301	2.473	2.530	2.497	
	Ebn at Noon	876	913	913	876	846	837	809	808	834	860	864	853	
	Edh at Noon	80	94	107	134	142	141	151	143	119	90	75	73	
All-Sky Solar Radiation	RadAvg	2.04	2.76	3.74	4.94	5.55	6.33	6.21	5.55	4.75	3.41	2.31	1.72	
	RadStd	0.23	0.28	0.35	0.29	0.48	0.49	0.43	0.32	0.30	0.38	0.24	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	+108
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80

Nomenclature: See separate page